



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
POWER SYSTEM OPERATION CORPORATION LIMITED
पावर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड
(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 22nd August 2018

To,

1. कार्यकारी निदेशक, पू. क्षेत्र. भा. प्रे. के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
2. महाप्रबंधक, ऊ. क्षेत्र. भा. प्रे. के., 18/ ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
General Manager, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
3. कार्यकारी निदेशक, प. क्षेत्र. भा. प्रे. के., एफ3-, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
4. कार्यकारी निदेशक, ऊ. पू. क्षेत्र. भा. प्रे. के., डोंगतेह, लोअर नोंग्रह, लापलंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
5. महाप्रबंधक, द. क्षेत्र. भा. प्रे. के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
General Manager, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 21.08.2018.

महोदय/Dear Sir,

आई०ई०जी०सी०-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 21-अगस्त-2018 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रा०भा०प्रे०के० की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 21st August 2018, is available at the NLDC website.

धन्यवाद,

पॉवर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड
राष्ट्रीय भार प्रेषण केंद्र, नई दिल्ली

Report for previous day

Date of Reporting

22-Aug-18

A. Maximum Demand

	NR	WR	SR	ER	NER	Total
Demand Met during Evening Peak hrs(MW) (at 2000 hrs; from RLDCs)	57304	42795	40172	21577	2704	164552
Peak Shortage (MW)	652	0	0	100	153	905
Energy Met (MU)	1283	967	874	453	52	3629
Hydro Gen (MU)	358	22	169	109	21	679
Wind Gen (MU)	20	102	226	-----	-----	348
Solar Gen (MU)*	16.22	7.72	48.73	0.62	0.01	73
Energy Shortage (MU)	10.4	0.0	0.0	0.3	1.0	11.7
Maximum Demand Met during the day (MW) & time (from NLDC SCADA)	59256	44317	39856	21975	2662	166331
	22:31	19:22	19:34	21:17	19:53	19:45

B. Frequency Profile (%)

Region	FV1	<49.7	49.7-49.8	49.8-49.9	<49.9	49.9-50.05	> 50.05
All India	0.032	0.00	0.41	6.82	7.22	81.52	11.26

C. Power Supply Position in States

Region	States	Max. Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	10512	0	237.1	142.6	-0.2	121	0.0
	Haryana	9280	0	194.1	150.8	0.8	161	0.0
	Rajasthan	9745	0	221.4	78.6	0.9	518	0.0
	Delhi	5852	21	119.4	98.0	-1.4	226	0.1
	UP	18821	0	389.3	192.0	0.8	338	0.0
	Uttarakhand	2096	0	43.1	22.9	0.0	105	0.0
	HP	1338	0	28.6	-1.8	2.9	245	0.0
	J&K	2246	562	43.6	19.3	-0.9	138	10.3
WR	Chandigarh	586	0	5.9	6.8	-0.9	-7	0.0
	Chhattisgarh	3574	0	81.8	16.7	-1.2	357	0.0
	Gujarat	13312	0	300.5	76.4	3.5	602	0.0
	MP	7282	0	160.2	74.4	-2.0	432	0.0
	Maharashtra	17455	0	379.8	104.0	-2.3	597	0.0
	Goa	441	0	9.0	8.8	-0.4	60	0.0
	DD	339	0	7.6	6.7	1.0	88	0.0
	DNH	779	0	18.2	17.1	1.1	86	0.0
SR	Essar steel	515	0	10.2	9.6	0.6	286	0.0
	Andhra Pradesh	7162	0	164.8	6.0	1.1	720	0.0
	Telangana	7570	0	150.0	41.6	-1.7	353	0.0
	Karnataka	8807	0	176.5	37.8	-2.0	600	0.0
	Kerala	3099	0	57.8	26.1	-0.5	231	0.0
	Tamil Nadu	14984	0	317.0	92.9	-1.0	399	0.0
	Pondy	381	0	8.1	8.0	0.1	85	0.0
ER	Bihar	4899	0	93.9	94.7	-1.9	200	0.0
	DVC	2958	0	65.9	-30.0	3.8	225	0.3
	Jharkhand	1183	0	23.8	18.4	-0.8	80	0.0
	Odisha	5122	0	100.6	51.0	3.0	225	0.0
	West Bengal	8245	0	167.6	50.2	1.9	220	0.0
	Sikkim	93	0	1.2	1.4	-0.2	12	0.0
NER	Arunachal Pradesh	123	1	2.2	2.5	-0.3	25	0.0
	Assam	1719	123	33.7	28.8	1.2	188	0.9
	Manipur	150	2	2.1	2.3	-0.2	25	0.0
	Meghalaya	294	0	5.1	1.0	-0.1	53	0.0
	Mizoram	78	0	1.5	1.0	0.1	17	0.0
	Nagaland	115	1	2.2	1.8	0.0	35	0.0
	Tripura	264	7	4.8	4.3	-0.5	103	0.1

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh
Actual(MU)	32.8	-7.5	-14.9
Day peak (MW)	1597.2	-302.0	-658.6

E. Import/export By Regions(in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	284.1	-238.4	2.9	-48.9	2.0	1.7
Actual(MU)	278.1	-236.4	-6.4	-35.6	0.9	0.7
O/D/U/D(MU)	-6.0	2.0	-9.3	13.3	-1.1	-1.1

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	Total
Central Sector	3751	15595	10262	1470	116	31194
State Sector	8220	17198	12703	5375	50	43546
Total	11971	32793	22965	6845	166	74740

G. Sourcewise generation (MU)

	NR	WR	SR	ER	NER	All India
Thermal (Coal & Lignite)	582	1030	361	405	8	2386
Hydro	358	22	169	109	21	679
Nuclear	27	27	24	0	0	79
Gas, Naptha & Diesel	20	39	23	0	25	108
RES (Wind, Solar, Biomass & Others)	50	110	311	1	0	472
Total	1038	1229	888	514	54	3724

Share of RES in total generation (%)	4.81	8.92	35.07	0.12	0.04	12.67
Share of Non-fossil fuel (Hydro, Nuclear and RES) in total generation (%)	41.94	12.96	56.79	21.28	38.64	33.02

H. Diversity Factor

All India Demand Diversity Factor	1.010
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Diversity factor = Sum of regional maximum demands / All India maximum demand

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

INTER-REGIONAL EXCHANGES							Date of Reporting : 22-Aug-18	
							Import=(+ve) /Export =(-ve) for NET (MU)	
Sl No	Voltage Level	Line Details	Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	765kV	GAYA-VARANASI	D/C	0	363	0.0	5.8	-5.8
2		SASARAM-FATEHPUR	S/C	346	0	5.1	0.0	5.1
3		GAYA-BALIA	S/C	0	377	0.0	6.2	-6.2
4		ALIPURDUAR-AGRA	-	0	823	0.0	20.8	-20.8
5	400 kV	PUSAULI B/B	S/C	0	398	0.0	9.7	-9.7
6		PUSAULI-VARANASI	S/C	0	297	0.0	0.0	0.0
7		PUSAULI -ALLAHABAD	S/C	0	163	0.0	0.0	0.0
8		MUZAFFARPUR-GORAKHPUR	D/C	0	632	0.0	10.3	-10.3
9		PATNA-BALIA	Q/C	0	1082	0.0	19.9	-19.9
10		BIHARSHARIFF-BALIA	D/C	0	189	0.0	3.3	-3.3
11		MOTIHARI-GORAKHPUR	D/C	119	0	6.2	0.0	6.2
12		BIHARSHARIFF-VARANASI	D/C	25	0	0.6	0.0	0.6
13	220 kV	PUSAULI-SAHUPURI	S/C	0	176	0.0	3.7	-3.7
14	132 kV	SONE NAGAR-RIHAND	S/C	0	0	0.0	0.0	0.0
15		GARWAH-RIHAND	S/C	35	0	0.6	0.0	0.6
16		KARMANASA-SAHUPURI	S/C	0	0	0.0	0.0	0.0
17		KARMANASA-CHANDAUJI	S/C	0	0	0.0	0.0	0.0
ER-NR						12.4	79.7	-67.3
Import/Export of ER (With WR)								
18	765 kV	JHARSUGUDA-DHARAMJAIGARH S/C	D/C	1020	2	16.5	0.0	16.5
19		NEW RANCHI-DHARAMJAIGARH	D/C	782	0	13.6	0.0	13.6
20	400 kV	JHARSUGUDA-RAIGARH	Q/C	1140	0	20.0	0.0	20.0
21		RANCHI-SIPAT	D/C	417	0	7.5	0.0	7.5
22	220 kV	BUDHIPADAR-RAIGARH	S/C	191	0	2.0	0.0	2.0
23		BUDHIPADAR-KORBA	D/C	210	0	3.6	0.0	3.6
ER-WR						63.1	0.0	63.1
Import/Export of ER (With SR)								
24	765 kV	ANGUL-SRIKAKULAM	D/C	17.0	855.9	0.0	8.1	-8.1
25	HVDC	JEYPORE-GAZUWAKA B/B	D/C	0.0	427.7	0.0	8.7	-8.7
26	LINK	TALCHER-KOLAR BIPOLE	D/C	0.0	992.6	0.0	19.3	-19.3
27	400 kV	TALCHER-I/C	D/C	947.9	0.6	0.0	18.2	-18.2
28	220 kV	BALIMELA-UPPER-SILERRU	S/C	0.0	0.0	0.0	0.0	0.0
ER-SR						0.0	36.0	-36.0
Import/Export of ER (With NER)								
29	400 kV	BINAGURI-BONGAIGAON	D/C	0	545	0.0	10.8	-11
30		ALIPURDUAR-BONGAIGAON	D/C	63	218	0.0	2.7	-3
31	220 kV	ALIPURDUAR-SALAKATI	D/C	0	124	0.0	2.3	-2
ER-NER						0.0	15.8	-15.8
Import/Export of NER (With NR)								
32	HVDC	BISWANATH CHARIALI-AGRA	-	0	701	0.0	17.1	-17.1
NER-NR						0.0	17.1	-17.1
Import/Export of WR (With NR)								
33	HVDC	CHAMPA-KURUKSHETRA	D/C	0	2004	0.0	47.8	-47.8
34		V'CHAL B/B	D/C	240	0	6.0	0.0	6.0
35		APL -MHG	D/C	0	1455	0.0	36.3	-36.3
36		GWALIOR-AGRA	D/C	0	1320	0.0	47.4	-47.4
37	765 kV	PHAGI-GWALIOR	D/C	0	1414	0.0	23.6	-23.6
38		JABALPUR-ORAI	D/C	0	705	0.0	23.7	-23.7
39		GWALIOR-ORAI	S/C	379	0	7.8	0.0	7.8
40		SATNA-ORAI	S/C	0	1778	0.0	41.8	-41.8
41	400 kV	ZERDA-KANKROLI	S/C	191	139	0.9	0.0	0.9
42		ZERDA -BHINMAL	S/C	98	199	0.0	1.3	-1.3
43		V'CHAL -RIHAND	S/C	963	0	21.5	0.0	21.5
44		RAPP-SHUJALPUR	D/C	0	399	0	3	-3
45	220 kV	BADOD-KOTA	S/C	24	56	0.2	0.4	-0.2
46		BADOD-MORAK	S/C	3	70	0.0	0.8	-0.8
47		MEHGAON-AURAIYA	S/C	44	17	0.3	0.0	0.3
48		MALANPUR-AURAIYA	S/C	18	34	0.0	0.4	-0.3
49	132kV	GWALIOR-SAWAI MADHOPUR	S/C	0	0	0.0	0.0	0.0
WR-NR						36.6	226.2	-189.5
Import/Export of WR (With SR)								
50	HVDC	BHADRAWATI B/B	-	0	309	0.0	7.2	-7.2
51	LINK	BARSUR-L.SILERU	-	0	0	0.0	0.0	0.0
52	765 kV	SOLAPUR-RAICHUR	D/C	1129	329	14.4	0.0	14.4
53		WARDHA-NIZAMABAD	D/C	342	843	0.0	2.4	-2.4
54	400 kV	KOLHAPUR-KUDGI	D/C	804	0	13.8	0.0	13.8
55		KOLHAPUR-CHIKODI	D/C	0	0	0.0	0.0	0.0
56	220 kV	PONDA-AMBEWADI	S/C	1	0	0.0	0.0	0.0
57		XELDEM-AMBEWADI	S/C	0	105	0.0	1.9	-1.9
WR-SR						28.2	11.5	16.7
TRANSNATIONAL EXCHANGE								
58		BHUTAN						32.8
59		NEPAL						-7.5
60		BANGLADESH						-14.9

POWER SYSTEM OPERATION CORPORATION LIMITED

Grid Monitoring Report for Kerala

22-Aug-2018 06:00 Hrs

1. Power Supply Position -

Description	Normal Value	12:00 Hrs (19 th Aug)	18:00 Hrs (19 th Aug)	24:00 Hrs (19 th Aug)	06:00 Hrs (20 th Aug)	12:00 Hrs (20 th Aug)	18:00 Hrs (20 th Aug)	24:00 Hrs (20 th Aug)
SR Grid Demand Met	37000-40000	34750	32932	30799	32253	34979	35429	32166
Kerala Demand Met	2500-2900	1823	1943	1793	1796	2264	2292	2039
Grid Frequency (Hz)	50.00	50.05	50.06	50.10	50.06	50.07	50.03	50.09
Kerala Hydro	1100-1300	1095	1098	1266	1256	1258	1295	1270
Drawl from Grid	1400-1600	682	808	545	523	951	1010	761

Description	Normal Value	06:00 Hrs (21 st Aug)	12:00 Hrs (21 st Aug)	18:00 Hrs (21 st Aug)	24:00 Hrs (21 st Aug)	06:00 Hrs (22 nd Aug)
SR Grid Demand Met	37000-40000	32937	37304	36341	35237	35389
Kerala Demand Met	2500-2900	1972	2403	2482	2322	2229
Grid Frequency (Hz)	50.00	50.07	49.99	50.12	50.03	50.02
Kerala Hydro	1100-1300	1238	1157	1279	1335	1297
Drawl from Grid	1400-1600	669	1238	1211	941	903

2. Generation / Demand particulars for Kerala

Parameter	Normal value	18 th Aug'18	19 th Aug'18	20 th Aug 18	21 st Aug 18
Kerala Peak Demand (MW)	3600	2505	2595	3002	3126
Kerala Energy Met (MUs)	62	44.5	45.1	53.8	57.76

3. Inter-State Lines Status

Name of Transmission Line	Status
400 KV Udumalpet - Palakkad D/C	In Service
400 kV Mysore - Kozhikode D/C	In Service
400 kV Tirunelveli - Trivandrum D/C	In Service
220 kV Idukki - Udumalpet	In Service
220 kV Edamon - Tirunelveli D/C	In Service
220 kV Kadakola - Kaniyampettai S/C	In Service
220 kV Theni - Sabarigiri	In Service

4. Generating Stations affected: (Details as per Annex-1)

Total Number of Generating stations affected due to flood (capacity in MW)	Total number of Generating stations Revived till Now (capacity in MW)	Generating units revived in last 06 hours (capacity in MW)	Remarks
17 (680MW)	5 (318 MW)	NIL	Total Number of Generating stations affected due to flood was updated as per latest report received from SLDC, Kerala

5. Substations Affected due to floods (Details as per Annex-1)

Voltage Level	Total Number of Substation affected due to flood	Total number of Substations Revived till Now	Substations revived in last 6 hours	Remarks
110 kV	06	04	Nil	110 KV Parappanangad Substation restored by 16:41 hrs of 20.08.18. 66kV Vaiga Threads restored at 19:46Hrs on 19.08.2018. 33 kV Vellikulangara S/s restored at 16:39 Hrs on 20.08.2018.
66 kV	05	03	Nil	
33 kV	06	04	Nil	

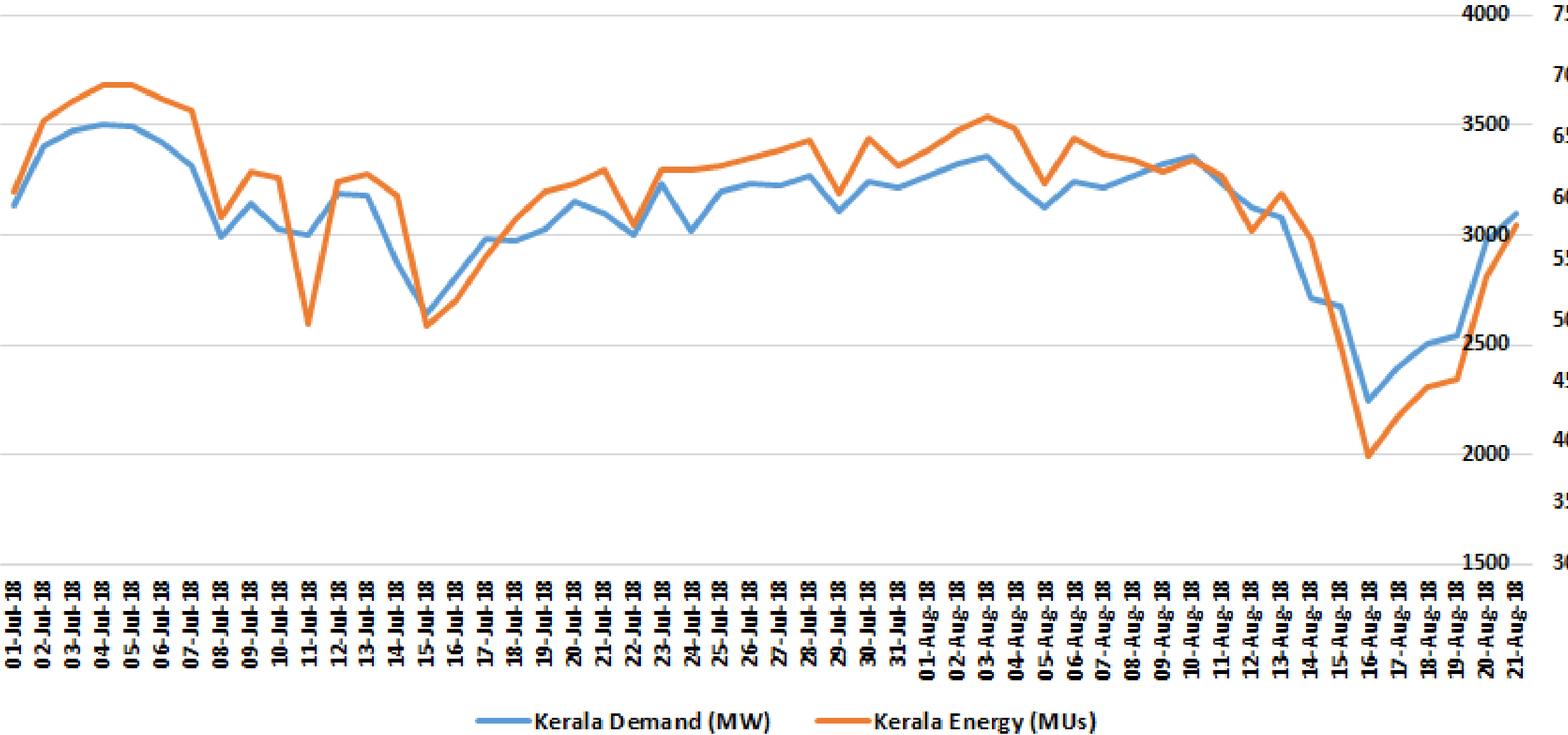
6. Transmission Lines Affected due to floods (Details as per Annex-1)

Voltage Level	Total Number of Transmission Lines affected due to flood	Total number of Transmission lines revived till now	Transmission lines revived in last 6 hours	Remarks
220 kV	03	03	Nil	
110 kV	14	09	Nil	01 no. 110 kV line restored at 11:42 Hrs on 21.08.18
66 kV	03	02	01	01 no. 66kV line restored at 12:36 Hrs on 21.08.18

Peak Demand Met

Kerala Demand Trend

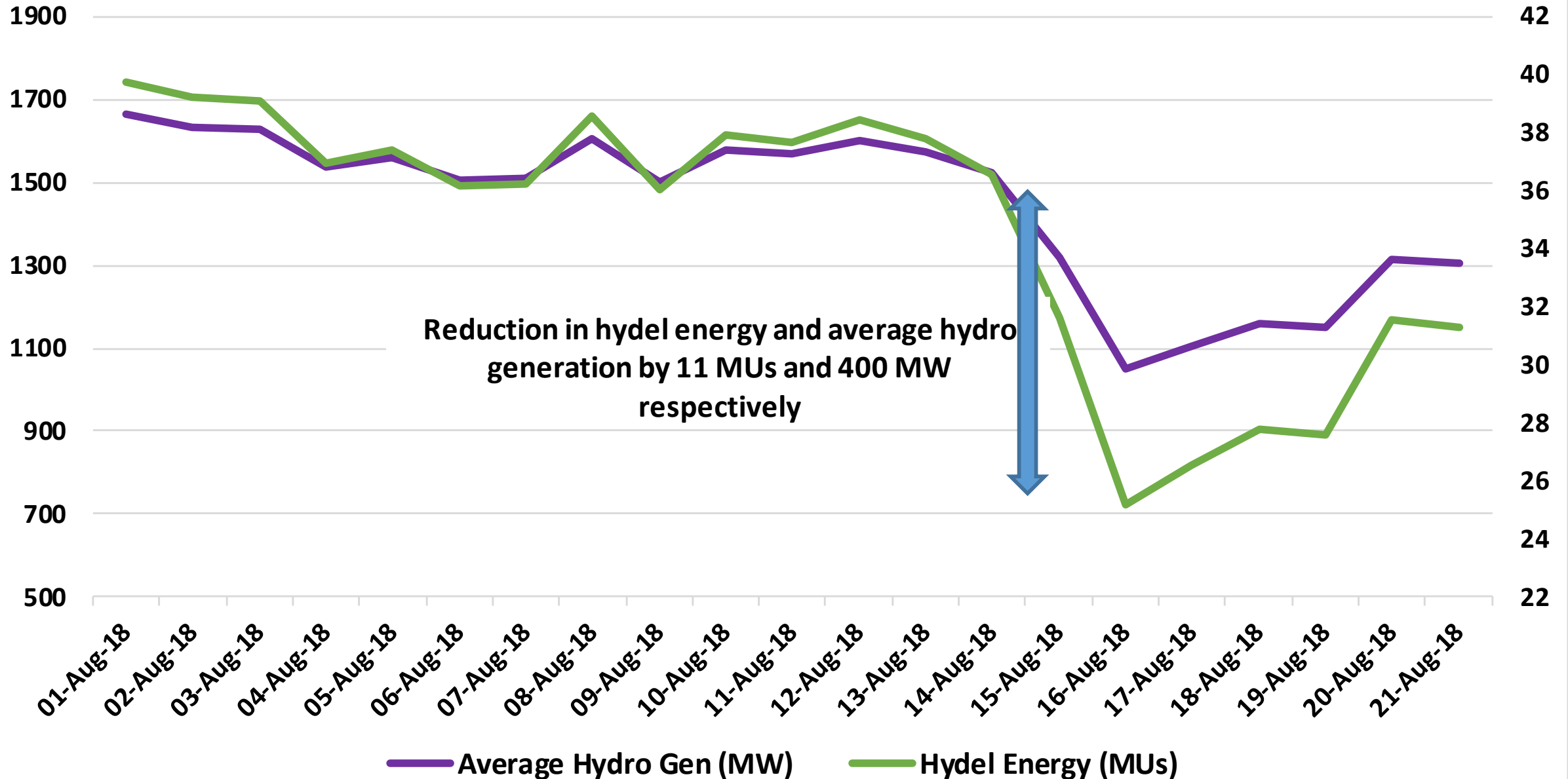
Energy Met



Kerala Hydro Generation

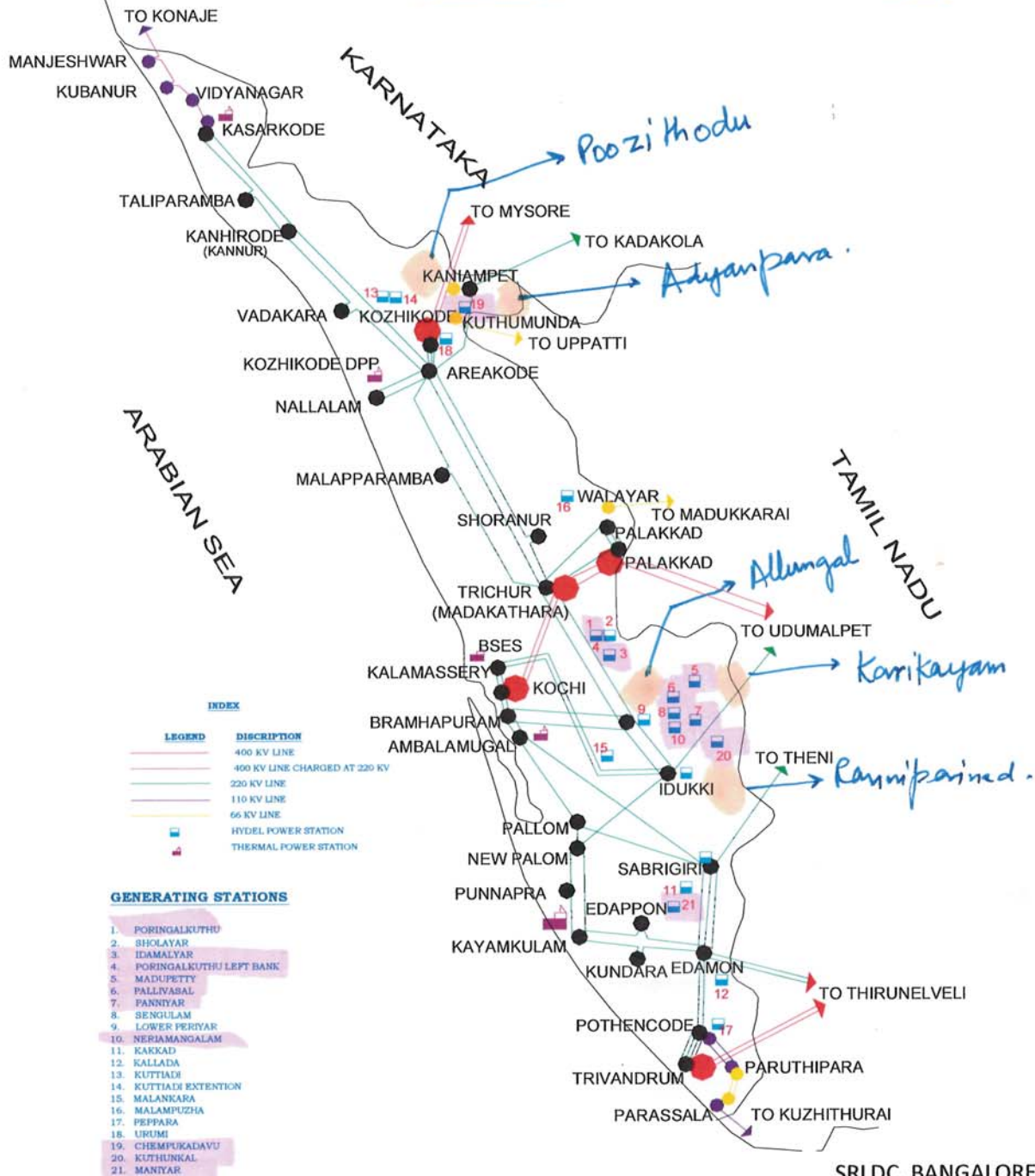
Ave. Hydro Gen.

Hydel Energy



POWER MAP OF KERALA

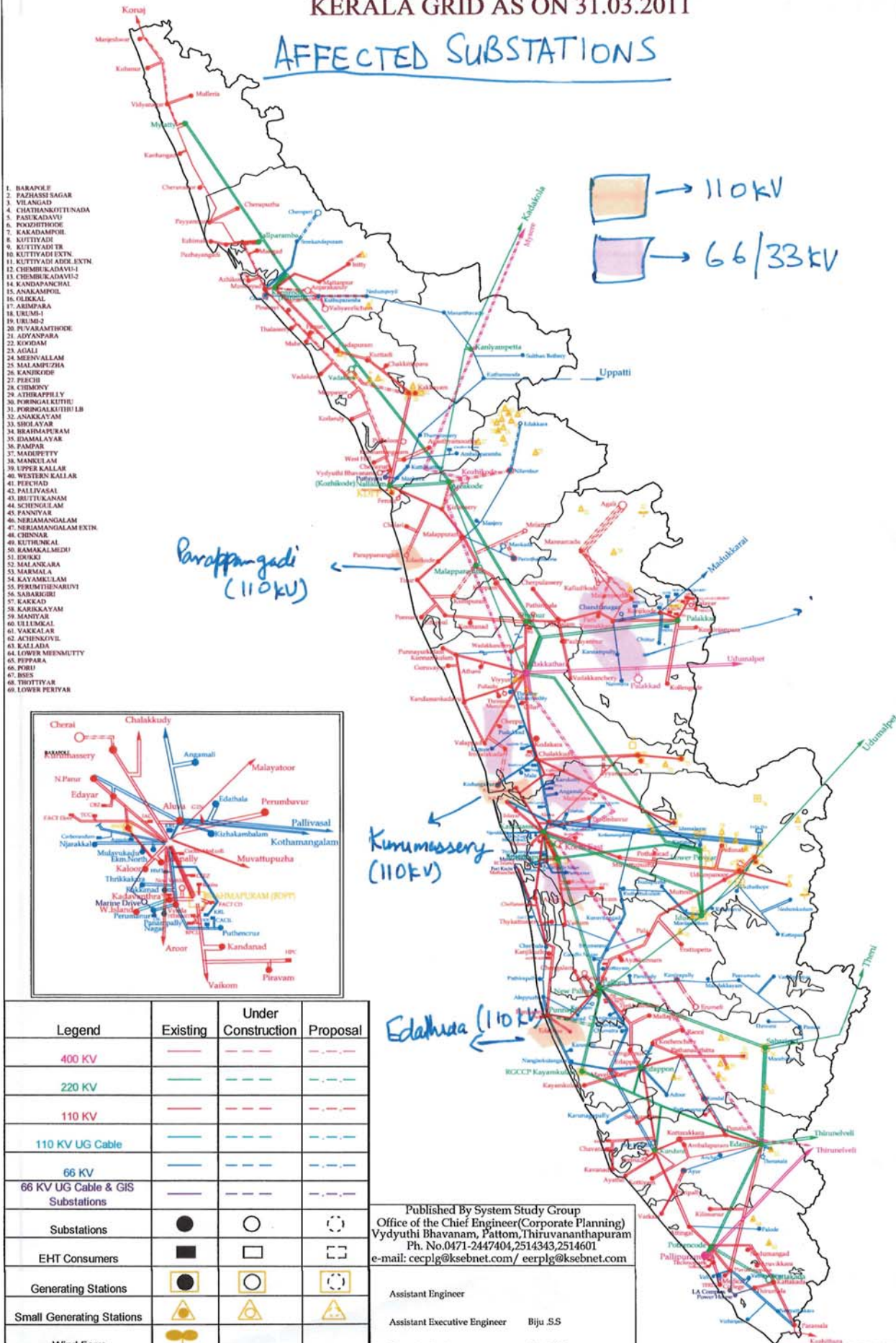
AFFECTED GENERATING STATIONS



SRLDC, BANGALORE
June 2018

KERALA GRID AS ON 31.03.2011

 $\rightarrow 110 \text{ kV}$
 $\rightarrow 66/33 \text{ kV}$



K. VALSAKUMARI
Chief Engineer
(Corporate Planning)

NATURAL CALAMITY- POWER SYSTEM CONDITION

Date: 22/08/2018 06:00 HRS

Generation Affected

	Sl.No.	Generating Station	Installed Capacity (MW)	Capacity affected due to inundation (MW)	Capacity revived till now (MW)	Present affected capacity (MW)	Reason for outage	Remarks
a	Due to transmission constraints							
	1	Idamalayar (2x37.5)	75	75	75	0	Line Constraints	Unit#1 revived at 23.50hrs on 19.08.18, Unit#2 revived at 00.39hrs on 20.08.18.
	2	Kuthungal (3x7)	21	21	0	21	Line constraints, Flooding of PH	outage since 15:03 hrs, 15.08.2018. Communication not yet restored.
	Sub-Total		96	96	75	21		
b	Due to power house inundation/tailrace water level high							
	1	Panniyar (2x16)	32	32	0	32	Land Sliding & water entered to control room.	outage since 23:49 hrs, 15.08.2018
	2	Neriamangalam (3x17.5 +25)	77.5	76	59	17	Partially affected (2X17.5)	Unit#1 switched off at 02:17 hrs 16.08.2018 ,due to cooling water outlet temperature high and revived at 15:52Hrs on 20.08.2018. unit #2 mechanical lockout at 00:27 hrs 16.08.2018. Unit#3 - Outage from 16:16 hrs on 16.08.2018 and restored at 13:01Hrs on 17.08.2018. Neriya mangalam Extension Scheme (NES) synchronized back at 13.38 hrs 18.08.2018
	3	Lower Periyar (3x60)	180	170	0	170	Due to flooding	Outage since 14:19 hrs ,14.08.2017, EHT Feeders charged through backfeeding, machines not running yet.
	4	Poringal (4x9 +16)	54	54	0	54	Due to flooding	Outage since 05:49 hrs, 16.08.2018
	5	Sabarigiri (2x60 +4x55)	340	170	170	0	Partially affected (2X55 + 60)	five machines in service from 19.30hrs. 55 MW machine is out due to other reasons prior to flood and continued to be kept under outage.
	6	Maniyar (3x4)	12	12	0	12	due to flooding	No generation & communication
	7	Allungal (2x3.5)	7	7	0	7	due to flooding	Outage since 22:45 hrs, 14-8-2018
	8	Karikayam (3x5)	15	10	10	0	due to flooding.	No2& No3 were under outage due to block in penstock path due to mud. Both units are revived by 16:00Hrs on 20.08.18
	9	Urumi 1 & 2 (3x1.25 + 3x0.8)	6.15	6.15	3.75	2.4	Canal Blocked due to land slide	Urumi-2 under Shutdown. Urumi-1 normalized at 13:00 Hrs on 20.08.2018
	10	Poozhithodu (3x1.6)	4.8	4.8	0	4.8	due to flooding	Outage since 22:05hrs, 8-8-2018
	11	Ranni-perinad (2x2)	4	4	0	4	due to flooding	Outage since 22:05hrs, 14-8-2018
	12	Chempukadavu (2.70+3.75)	6.45	6.45	0	6.45	due to flooding	Outage since 17:17hrs, 15-8-2018
	13	MATTUPETTY (2)	2	2	0	2	due to flooding	Outage since 15.00hrs, 11-8-2018
	14	ADYANPARA (2x1.5 +0.5)	3.5	3.5	0	3.5	due to flooding	Outage since 14-08-2018
	Sub-Total		744.4	557.9	242.75	315.15		
c.	Due to other reasons/Flood control							
	1	Pallivasal (3x7.5+3x5)	37.5	26	0	26	For flood control in Munnar Town	Outage since 12:23 hrs,14.08.2018
	Sub-Total		37.5	26	0	26		
Grand Total			878	680	318	362		

Transmission System Affected

Substation switched off due to inundation					
Sl.No.	Substation Name	Reason for Outage			Remarks
1	110KV KURUMASSERY	WATER ENTERED TO CR			switched off at 17:26 hrs 15.08.2018-
2	110KV MALAYATTOOR	WATER ENTERED TO CR			All EHT feeders restored by 23.00hrs on 19.08.18
3	110KV EDATHUA	WATER ENTERED TO CR			switched off at 21:35 hrs 16.08.2018
4	110 KV KANDASANKADAV	WATER ENTERED TO CR			Restored by 9.45Hrs, 18-8-2018
5	110KV PARAPPANANGAD	WATER ENTERED TO CR			switched off at 04:32 hrs, 18.08.2018. 110kv feeder restored at 11:02hr on 20.8.18;Transformer#2-charged-13:39Hrs on 20.8.18;Transfrmr#1 charged at 16:41Hrs on 20.8.18
6	110 KV EDAYAR	WATER ENTERED TO CR			switched off at 12:43 hrs,16-8-2018. restored by 16.15hrs on 19.08.18
7	66 KV KCPL	WATER ENTERED TO CR			switched off at 8:55 hrs, 16-8-2018
8	66 KV KUTTANAD	WATER ENTERED TO CR			switched off at 10.35 Hrs, 17-8-2018
9	66 KV VAIGA THREADS	NON AVAILABILITY OF FEEDER			switched off at 12.01 Hrs, 18-8-2018. Restored at 19:46Hrs on 19.08.2018
10	66KV KORATTY	WATER ENTERED TO CR			switched off at 12.01Hrs, 18-8-2018 & Restored by 12.50Hrs on 19.08.18
11	66KV MULAVUKAD	WATER ENTERED TO CR			Trfr #1 charged at 17.09Hrs, trfr #2 charged at 17.16Hrs on 19.08.18
12	33 KV VELLIKULANGARA	WATER ENTERED TO CR			switched off at 12.01Hrs, 15-8-2018. Restored at 16:39Hrs on 20.08.2018
13	33 KV KALLETUMKARA	WATER ENTERED TO CR			Restored at 18.15 hrs on 19.08.18
14	33 KV VADAKEKARA	WATER ENTERED TO CR			switched off at 12.44Hrs, 16-8-2018
15	33 KV VARAPUZHA	WATER ENTERED TO CR			Restored at 14.25 hrs on 19.08.18
16	33 KV ALANGAD	WATER ENTERED TO CR			switched off at 12.46 Hrs, 16-8-2018
17	33 KV THAKAZHI	WATER ENTERED TO CR			Restored at 15.31hrs , 19-8-2018

2	b	EHT lines(66kV and above) s/off due to flood related issues			
		Sl.No.	Name of Transmission Line	Reason for Outage	Remarks
		1	220 KV Madakathra-Lower Periyar I&II	Lines switched off due to landsliding at Lower Periyar Substation	Restored at 14.28Hrs, 16-8-2018
		2	220 KV Brahmapuram-Lower Periyar I&II		Restored at 10:58hrs, 19-8-2018
		3	220 kv IDUKKI-LOWER PERIYAR I&II		Restored at 14.28hrs, 16-8-2018
		4	110KV PUNNAPRA - CHENGALA	FOR SAFETY CLEARANCE	switched off at 19:06 hrs 17.08.2018
		5	110KV PUNNAPRA - KODIMATHA	FOR SAFETY CLEARANCE	Switched off at 19:06 hrs 17.08.2018
		7	110KV CHENGANNUR- TRIKODITHANAM	FOR SAFETY CLEARANCE	Switched off at 13:04 hrs 16.08.2018 and restored at 15:10Hrs on 20.08.2018
		8	110KV CHENGANNUR- MALLAPPILLY	FOR SAFETY CLEARANCE	Switched off at 13:20 hrs 16.08.2018 and restored at 16:11Hrs on 20.8.2018
		9	110KV PUNNAPRA-EDATHUA #1,#2	FOR SAFETY CLEARANCE	Switched off at 21:35 hrs 16.08.2018
		10	110KV KALAMASSERY- EDAYAR#1,#2	DUE TO FLOODING	110KV Kalamassery Edayar #2 charged at IAC at 11:23 hrs 17.08.2018,Ckt#1 charged at 10.51Hrs on 19-8-2018
		11	110 KV KALAMASSERY-METRO #1,2	FOR SAFETY CLEARANCE	switched off at 7:12 hrs 16.08.2018. alternate supply availed at 33kv level
		12	110KV CHELARI-PARAPPANANGADI	FOR SAFETY CLEARANCE	Switched off at 04:32 hrs 18.08.2018 and restored on 20.08.18 at 11.02 hrs.
		13	110 KV ALUVA-CARBORANDAM-KALAMASSERY	DUE TO ACCESS FAILURE	TRIPPED DUE TO TREE FALL at 21:31Hrs on 15-8-2018. Restored at 20.08.2018 at 18:31 Hrs
		14	110KVPORINGAL-CHALAKUDY1	DUE TO FLOODING	Switched off at 22:19hrs, 15-8-2018
		15	110KV PORINGAL-CHALAKUDY2	DUE TO FLOODING	Switched off at 5:50hrs, 16-8-2018 and restored at 12:49 Hrs on 20.8.18
		16	110 KV CHALAKUDY-TRACTION	DUE TO FLOODING	Tripped at 15:07 hrs, 16-8-2018. Revived at 11:24 hrs on 21.8.18
		17	110 KV KALAMASSERY-EDAPILLY	DUE TO FLOODING	Switched off at 17:14hrs, 16-8-2018 and restored on 20.08.18 at 08.36 hrs
		18	110 KV EDAYAR-NORTH PARAVUR 1&2	DUE TO FLOODING	Switched off at 12:43hrs ,Ckt#1 & 2 restored at 11.30Hrs &11.52Hrs respectively on 19-8-2018
		19	66 KV ALUVA-PALLIVASAL-1	DUE TO ACCESS FAILURE	TRIPPED DUE TO TREE FALL 15-8-2018, 16:53hrs. Restored at 12:36 hrs on 21.08.18
		20	66KV PALLOM - KUTTANAD	FOR SAFETY CLEARANCE	switched off at 11:08 hrs 17.08.2018
		21	66KV ALUVA - ANGAMALLY	DUE TO ACCESS FAILURE	switched off at 12:01 hrs, 18-8-2018. charged at 17.46 hrs on 19.08.18
3		Failure of transmission Lines			
		1	Neriamangalam-Kuthungal Line	Tower Slanting	switched off on 14.08.2018